Kexin Liu

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Education

Tulane University Aug 2024 - Present

Ph.D. Student in Biostatistics and Data Science | GPA: 3.96/4.0

Coursework: Statistical Inference, Bayesian Inference, Categorical Data Analysis, Clinical Trials, Epidemiologic Methods

Renmin University of China Sep 2021 - Jun 2024

M.S. in Epidemiology and Health Statistics | GPA: 3.84/4.0

Coursework: Survival Analysis, Causal Inference, Multivariate Analysis, GLMs, Statistical Computing

China Pharmaceutical University Sep 2017 - Jun 2021

B.S. in Marine Pharmacy | GPA: 3.3/4.0

Research Experience

Research Assistant, TRIAD - Tulane University School of Medicine

Foundation Model-Based LVEF Assessment from Portable ECGs - DECAAF II

- Developed a single-lead ECG pipeline combining frozen ECGFounder embeddings, repeated-recording pooling, multitask prediction, and clinical covariates to assess reduced LVEF after AF ablation.
- Implemented repeated patient-level nested cross-validation, cross-fitted stacking and calibration, and sensitivity analyses of rhythm, recording windows, and clinical covariates.
- Achieved internal AUROCs of 0.908 for LVEF <=40% and 0.857 for LVEF <50% in 359 participants; continuous LVEF MAE was 5.09 percentage points.
- Evaluated imaging-prioritization rules and decision curves; applied the locked pipeline exploratorily to 15 external patients and prepared a manuscript and HRS abstract.

Wearable PPG and Incident Heart Failure - Exploratory Analysis

- Linked longitudinal wearable PPG recordings with EHR follow-up and incident-HF dates; assessed case definitions, identifier linkage, pre-event coverage, and participant-level feasibility.
- Built reproducible signal-quality and pulse-interval feature extraction workflows spanning 25/100 Hz recordings, with pulse rate, robust interval variability, RMSSD, and sample entropy.
- Investigated timestamp artifacts and preprocessing sensitivity; designed within-person pre/post-HF comparisons while accounting for recording availability and short-segment measurement limits.

Spatial Transcriptomics of Atrial Fibrillation and Ablation

- Analyzed human left atrial posterior-wall Visium HD data across non-AF, AF, and ablated tissue using reference-based annotation, cell2location deconvolution, and multiresolution workflows.
- Characterized cardiomyocyte and fibroblast expression, spatially variable genes, pathway enrichment, and CellChat ligand-receptor signaling to investigate atrial remodeling.
- Investigated recovery-associated gene programs, cell-type colocalization, and fibrotic border-zone patterns; incorporated donor-level replication and tissue-composition limitations into interpretation.

MRI-ECG Cross-Modal Learning for Atrial Substrate - Exploratory Analysis

- Explored MRI-informed ECG representation learning and contrastive alignment for post-ablation fibrosis and left atrial remodeling tasks.
- Evaluated patient-held-out generalization, leakage, and patient-identity shortcuts to distinguish transferable substrate information from cohort-specific signal.

Selected Statistical Methods Research

Instrumental Variable Estimation with Semi-Competing Risks

- Developed a counting-process framework for causal effects on disease onset, death before onset, and post-onset death; formulated weighted entry balance to address selection into post-onset risk sets.
- Derived identification and large-sample results, implemented recursive estimators and simulation studies, and applied the framework to REACHnet comparative-safety data.

Composite Score Regression and Component Adjustment

- Established equivalence conditions, sensitivity bounds, and residual-variance diagnostics for interpreting a composite-score coefficient after component adjustment.
- Validated the framework in nine simulation designs (2,700 samples) and a REACHnet application involving 3,956 selected admissions; prepared a manuscript for epidemiologic audiences.

Neural Network Dynamic Prediction for Interval-Censored Data

- Integrated multivariate functional PCA and neural networks for dynamic risk prediction with time-varying covariates; conducted nonlinear and high-dimensional simulation studies.
- Applied the method to ADNI data (AUC 0.93; Brier score 0.024) and built a Shiny application for personalized prediction.

Publications

Neural Network-Based Dynamic Prediction for Interval-Censored Data With Time-Varying Covariates (2025)
Liu, K., Zu, Y., Yi, D., Ding, Y., Sun, T. doi: 10.1002/sim.70190

Relationships Between Fatigue Experience and Psychological Indicators Among Chinese Cancer Survivors (2023)
Gu, S., Xu, Y., Zhu, X., Lam, A., Gong, L., Wang, J., Guo, X., Fu, L., Shi, J., Liu, K., Yi, D. doi: 10.3389/fonc.2023.1194673

Manuscripts in Preparation

AI-enabled echocardiography triage of left ventricular ejection fraction from portable single-lead electrocardiograms in patients with atrial fibrillation: DECAAF II.

Instrumental variables estimation with semi-competing risks: a counting-process approach under weighted entry balance.

When Does Adjustment for a Composite-Score Component Remove Its Points? A Diagnostic Framework for Epidemiologic Regression.

Editorial Experience

Statistical Student Editor, American Journal of Public Health (AJPH)

- Evaluate statistical methods, study design, model assumptions, uncertainty reporting, and interpretation in submitted public-health manuscripts; provide methodological recommendations and assess responses to review.

Industry Experience

Machine Learning Intern, Deepwise - Beijing, China Oct 2022 - Jan 2023

- Reviewed AI methods for clinical imaging; preprocessed brain MRI images through pixel conversion and resampling; contributed to research writing and revision.

Data Analytics Intern, WuXi Biologics - Shanghai, China Jun 2022 - Sep 2022

- Preprocessed and annotated antibody datasets; built an annotation dictionary and trained a ProteinBERT-based sequence model, achieving 87% prediction accuracy.

Technical Skills

Programming and Tools: Python, R, SAS, Linux, Git, Shiny; SPSS.
Deep Learning and AI: PyTorch; neural networks and multitask learning; ECGFounder and Transformer-based ProteinBERT models; frozen-encoder feature extraction; cross-modal and contrastive representation learning.
Statistical Modeling: Survival and interval-censored data; semi-competing risks; instrumental variables and causal inference; functional PCA; GLMs; simulation and bootstrap inference.
Model Evaluation and Biomedical Data: Patient-level nested cross-validation, calibration, decision-curve analysis; ECG/PPG signal processing; MRI-ECG integration; spatial transcriptomics, cell2location, CellChat, bin2cell.
Languages: English, Mandarin, Spanish.